Modelling and Discovery of an Artificial Neuron for Quantitative Prediction in a Predator-Prey Ecosystem Dynamic

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Abstract

The description of population evolution among different species is essential to understanding the dynamics within an ecosystem. This is a complex and realistic subject, especially considering that these populations are not isolated. Mathematical models have been developed to describe these interactions and have achieved significant results, but predator-prey models have yet to reveal all their mysteries.

These models can be used to predict and manage animal populations in natural parks. For instance, in the event of a sharp increase in the number of predators, park management might consider authorizing the hunting of a certain number of animals to prevent excessively chaotic population dynamics-this is where the concept of parameter calibration becomes crucial.

Our approach incorporates this aspect along with other factors, such as the influence of hunting, to train an artificial neuron capable of predicting populations or animal numbers for future generations. Following the modeling and numerical simulation based on differential equations (Lotka-Volterra model), which illustrate the evolution of two distinct species' populations in an ecosystem, mathematical processing of the studied model and statistical methods were employed to optimize the input parameters of the artificial neuron.

Keywords and phrases: Predator-prey model,numerical simulation,differential equations,artificial neuronal network,population prediction

1 Introduction

The extraordinary diversity of life demands a delicate classification process: it applies to a dynamic system, whose elements evolve continuously. Mathematics offers rigorous tools that allow, among other things, the description of population dynamics and the understanding of prey-predator equilibria.

Since the works of Malthus, Verhulst, and their successors, various mathematical models have been developed to describe the evolution of an isolated population. However, reality is far more complex: multiple species often coexist within the same ecosystem, mutually influencing their respective populations.

In the 1920s, the Italian zoologist Umberto D'Ancona (1896-1964) studied fishing statistics in the Adriatic Sea. He observed that during World War I, a period of restricted fishing, the percentage of selachians (sharks, rays, etc.), considered unfit for consumption, was significantly higher than before the war (and than it would be after the war). Intrigued by this phenomenon, he asked his future father-in-law, mathematician Vito Volterra, in 1925 to develop a mathematical model to explain it. This led to the creation of the Lotka-Volterra model [1].

This model inspired numerous experimental studies. The Hudson Bay Company, for instance, analyzed hunting data collected between 1847 and 1903 on lynx and hare populations in Canada. These data confirmed the oscillations in the populations of the two species, consistent with the predictions of the Lotka-Volterra model.

Our approach combines these biological observations with modern modeling in artificial intelligence. We propose to overlay these observations with a probabilistic model based on a continuous probability distribution. This distribution will serve as an activation function in an artificial neuron designed to recognize prey-predator dynamics and predict their respective population sizes. Finally, statistical and optimization methods will be used to refine the neuron's input parameters and improve its predictions.

2 Biological Background and Mathematical Formulation of the Lotka-Volterra Model

2.1 Biological Background

The mathematical model proposed by Lotka-Volterra describes the evolution of two populations consisting of a predator and a prey occupying the same territory. For instance, lions and roosters can be taken as an example: lions feed on roosters, while the roosters have access to an unlimited supply of seeds. If $l(t)$ represents the size of the lion population at time t, and $c(t)$ represents the size of the rooster population, Volterra makes the following simplifying assumptions over a very short time interval from t to $t + \Delta t$:

- In the absence of roosters, the lion population follows a Malthusian model (it decreases), and its decline is proportional to Δt and the number $l(t)$ of lions.
- In the presence of roosters, the number Δl of lions increases proportionally to Δt, as well as to the number $l(t)$ of lions and the number $c(t)$ of roosters.
- In the absence of lions, the rooster population also follows a Malthusian model (it increases), and its growth Δc is proportional to Δt and the number $c(t)$ of roosters.
- The number Δc of roosters eaten by lions is proportional to Δt, the number $c(t)$ of roosters, and the number $l(t)$ of lions.

2.2 Mathematical Formulation of the Lotka-Volterra Model

The above assumptions are mathematically expressed as follows :

$$\begin{cases} \Delta l = [-al(t) + bc(t)l(t)]\,\Delta t, \\ \Delta c = [dc(t) - ec(t)l(t)]\,\Delta t. \end{cases}$$

Equivalently:

$$\begin{cases} \frac{\Delta l}{\Delta t} = -al(t) + bc(t)l(t), \\ \frac{\Delta c}{\Delta t} = dc(t) - ec(t)l(t), \end{cases}$$

where a, b, d, and e are positive constants. Thus, Volterra arrives at the system of differential equations:

$$\begin{cases} \frac{dl}{dt} = l(t)\left[-a + bc(t)\right], \\ \frac{dc}{dt} = c(t)\left[d - el(t)\right]. \end{cases}$$

This model allows Volterra to study the impact of hunting on such a population. If sufficiently zealous hunters differently decimate the two populations by reducing their growth rates by the amounts f_1 and f_2, respectively, the above differential equations become:

$$\begin{cases} \frac{dl}{dt} = l(t)\left[-(a + f_1) + bc(t)\right], \\ \frac{dc}{dt} = c(t)\left[(d - f_2) - el(t)\right]. \end{cases}$$

The parameters of the system are as follows:

- a: the natural growth rate of the prey in the absence of predators, in day^{-1},
- b: the predation rate, in $\text{day}^{-1}\text{N}^{-1}$,
- e: the mortality rate of the predators, in day^{-1},
- d: the reproductive efficiency of the predators based on the amount of prey captured, in $\text{day}^{-1}\text{N}^{-1}$.

3 Mathematical Treatment of the Volterra System

Let us revisit the system of differential equations derived by Volterra. By dividing the two equations of the model term by term, we can write:

$$\frac{[dc(t) - ec(t)l(t)]}{l(t)} = \frac{[-al(t) + bc(t)l(t)]}{c(t)}.$$

This leads to:

$$\left[\frac{d}{l(t)} - e\right] l'(t) = \left[\frac{-a}{c(t)} + b\right] c'(t).$$

By integrating, we obtain (denoting the natural logarithm as ln and K as a constant):

$$d\ln[l(t)] - el(t) = -a\ln[c(t)] + bc(t) + K.$$

Thus, the points remain on the curve (which is closed) described by the equation:

$$a\ln[c(t)] + d\ln[l(t)] = bc(t) + el(t) + K.$$

Thus, the points remain on the curve (which is closed) described by the equation:

$$a\ln[c(t)] + d\ln[l(t)] = bc(t) + el(t) + K, \quad \text{soit encore:}$$

$$\exp[bc(t) + el(t)] = C[c(t)]^a[l(t)]^d$$

According to Alexander Lyapunov's theory of stability for dynamical systems, by setting $l'(t)$ and $c'(t)$ to zero in the two equations of the model, we can determine the equilibrium point (other than the origin of the plane):

$$l_0 = \bar{l} = \frac{d - f_1}{e}, \quad c_0 = \bar{c} = \frac{a + f_2}{b}$$

$$\text{If } f_1 = 0 \quad \text{and} \quad f2 = 0$$

$$l_0 = \bar{l} = \frac{d}{e}, \quad c_0 = \bar{c} = \frac{a}{b}$$

If we perform a change of variables to translate the axes such that the origin of the plane coíncides with the equilibrium point, which is equivalent to setting:

$$l(t) = l_0 + x(t), \quad c(t) = c_0 + y(t),$$

We then arrive at the system:

$$\begin{cases} x'(t) = l'(t) = -a[l_0 + x(t)] + b[l_0 + x(t)][c_0 + y(t)] = \frac{bd}{e}y(t) + bx(t)y(t), \\ y'(t) = c'(t) = d[c_0 + y(t)] - e[l_0 + x(t)][c_0 + y(t)] = -\frac{ae}{b}x(t) - ex(t) - ex(t)y(t). \end{cases}$$

For very small displacements around the equilibrium point, we can neglect the product of $x(t)$ and $y(t)$, leading to the following new system:

$$\begin{cases} x'(t) = \frac{db}{e}y(t), \\ y'(t) = -\frac{ae}{b}x(t). \end{cases}$$

$$\text{Then: } \begin{cases} x''(t) = \frac{db}{e}y'(t), \\ y''(t) = -\frac{ae}{b}x'(t), \end{cases}$$

$$\text{and lastly } \begin{cases} x''(t) = -adx(t), \\ y''(t) = -ady(t). \end{cases}$$

These two second-order differential equations are analogous to those found in physics for the oscillations of a spring or pendulum. Their general solutions are given by:

$$x(t) = C_1 \cos(\omega t) + C_2 \sin(\omega t), \quad y(t) = C_3 \cos(\omega t) + C_4 \sin(\omega t),$$

where C_1, C_2, C_3, C_4 are constants, and $\omega = \sqrt{ad} > 0$. The solutions are therefore periodic, with a period:

$$T = \frac{2\pi}{\omega}.$$

By adopting specific initial conditions $x(0) = \epsilon$ and $y(0) = 0$, we obtain:

$$x(t) = \epsilon \cos(\omega t), \quad y(t) = -\frac{ae\epsilon}{b\omega} \sin(\omega t),$$

which which describes an ellipse:

$$\frac{(x(t))^2}{\epsilon^2} + (\frac{b\omega}{ae\epsilon})^2 (y(t))^2 = 1 \text{[2]}$$

Futhermore,

$$(ln[l(t)])' = \frac{l'(t)}{l(t)} = -a + bc(t) \quad \text{and} \quad (ln[c(t)])' = \frac{c'(t)}{c(t)} = d - el(t),$$

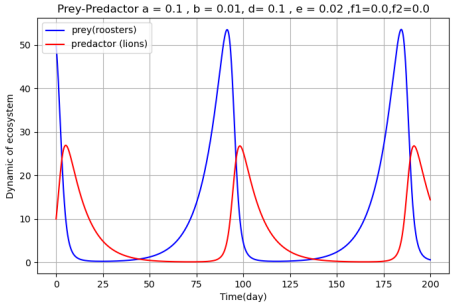
we obtain, by integrating over the period T so that $l(0) = l(T)$ and $c(0) = c(T)$:

$$\bar{l} = \frac{1}{T} \int_0^T l(t)dt = \frac{d}{e} \quad \text{and} \quad \bar{c} = \frac{1}{T} \int_0^T c(t)d(t) = \frac{a}{b}$$

Thus, the mean values coïncide with the equilibrium values.

4 Numerical Simulation

Herein, we present the results of numerical simulation of Lokta-Volterra system using scipy in python.We have superimposed in each case the essence of the phenomenon (the observed dynamics) with the density of the normal distribution with a mean equal to the average,$\mu = \bar{l}$ and volatility(variance) $\sigma^2 = \bar{c}$ for predactors and $\mu = \bar{c}$ and variance $\sigma^2 = \bar{l}$ for preys,to mathematically understand the phenomenon and observe the attraction probabilities (the region where the prey population tends to decrease while the predator population increases)[4]. All of this is summarized through the following simulations.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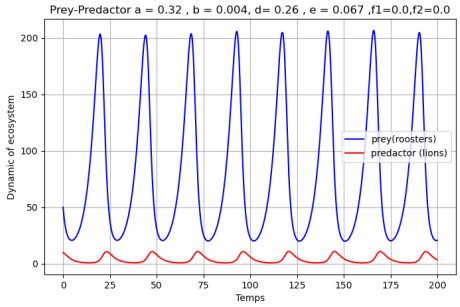
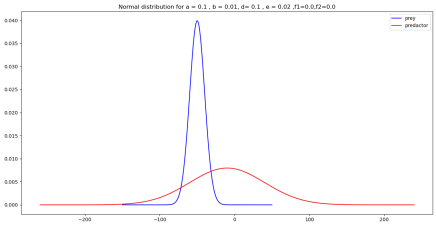
prey(roosters)
predactor (lions)
Dynamic of ecosystem
Time(day)


Prey-Predactor a = 0.32 , b = 0.004, d= 0.26 , e = 0.067 ,f1=0.0,f2=0.0
prey(roosters)
predactor (lions)
Dynamic of ecosystem
Temps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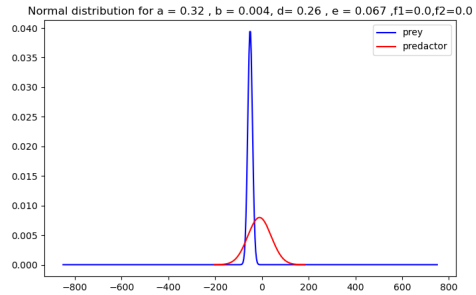
prey
predactor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prey
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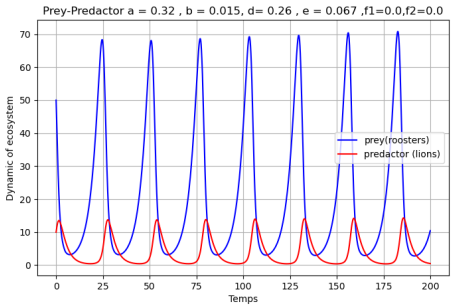
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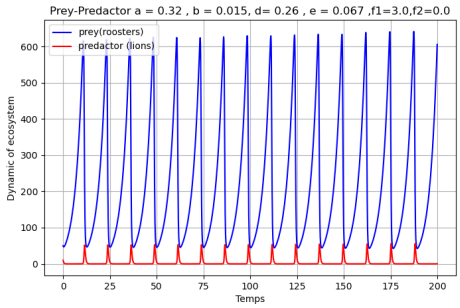
prey(roosters)
predactor (lions)
Dynamic of ecosystem
Temps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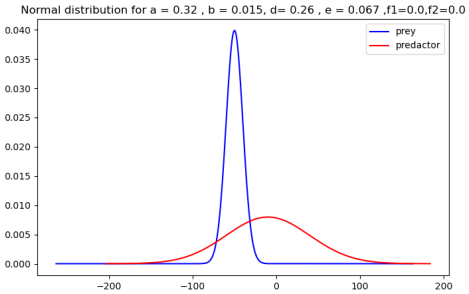
prey(roosters)
predactor (lions)
Dynamic of ecosystem
Temps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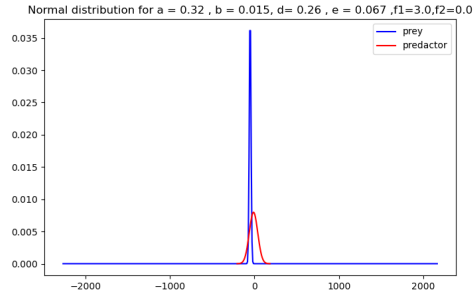
prey
predactor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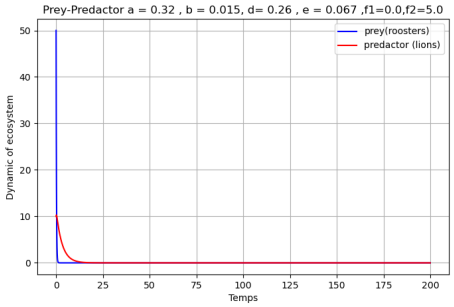
prey
predactor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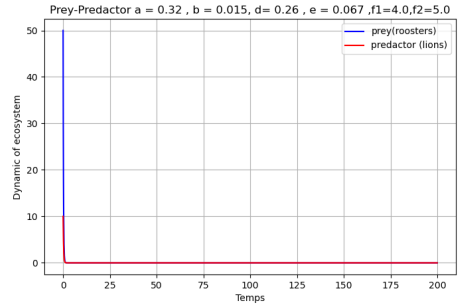
prey(roosters)
predactor (lions)
Dynamic of ecosystem
Temps


Prey-Predactor a = 0.32 , b = 0.015, d= 0.26 , e = 0.067 ,f1=4.0,f2=5.0
prey(roosters)
predactor (lions)
Dynamic of ecosystem
Temps

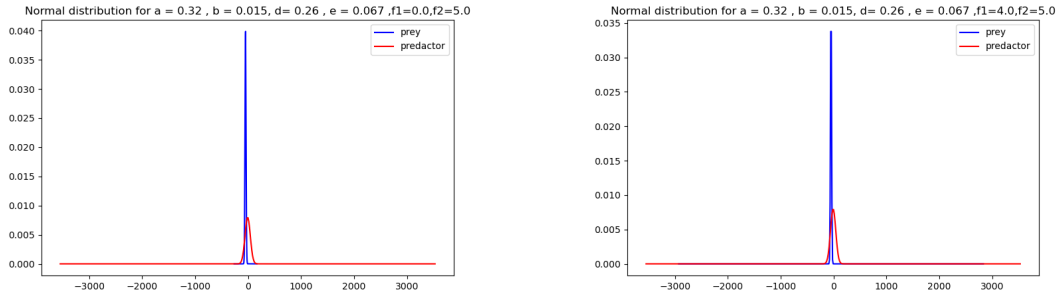


FIGURE 1: Numerical temporal simulation and superposition by normal distribution in each cases

One observation was the influence of hunting on the evolution of populations (parameters f_1 and f_2.) It was found that hunting on average increases the prey population and decreases on average the predator population.

5 Modelling of Artificial Neurone of Prediction

We use a linear model, the perceptron,since it is the model found at the core of neural networks.It is a binary model capable of linearly separating two classes of points. The operation of the neuron is modeled by a transfer function Z that takes l(t) and c(t) as inputs and returns an output y.The aggregation function is defined by $Z(l,c) = \omega_l l + \omega_c c$+b. ω_1 and ω_2 are called weigths and b the bias.The decision boundary corresponds to $Z(\omega_l, \omega_c) = 0$.The goal is to train an artificial neuron [7]on our reference data (l,c) so that it strengthens the parameters ω_l and ω_c each time an input is activated simultaneously with the output y present in this data.

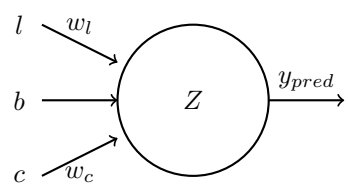


FIGURE 2: Modeling the Functioning of an Artificial Neuron [3]

$$\text{If} \begin{cases} z(l,c) < 0, y_{pred} = 0 \\ z(l,c) \geq 0, y_{pred} = 1 \end{cases}$$

Each prediction must be accompanied by a probability

5.1 Cost function in Machine Learning and errors minimization

The cost function is obtained by the Maximum Likelihood Method [4]and allows to quantify the errors commit by a model in Machine Learning. The probability of belonging to class 1 or the probability of being a predator follows a normal distribution with a given mean $\mu = \bar{l} = \frac{d-f_1}{e}$, and the variance $\sigma^2 = \bar{c} = \frac{a+f_2}{b}$,$N(\bar{l}, \bar{c})$ and given by:

$$a(Z) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{Z} \exp\left(-\frac{(l-\mu)^2}{2\sigma^2}\right) dl$$

.

a(Z) is the activation function.

Hebb's Theory (5.1.1): When two biological neurons are connected, they strengthen their synaptic links. Based on this theory, we will build backpropagation, which consists of determining how the network's output varies as a function of its parameters ω_l, ω_c.In the mathematical treatment of the Volterra System, we have $\omega = \sqrt{ad}$ then, we consider $\omega_l = \sqrt{ad}$ and $\omega_c = \sqrt{be}$ to ensure the equilibrium.

5.1.1 Maximun Likehood Method

The cost function is obtained using the maximum likelihood method and allows quantifying the errors made by a model in Machine Learning. The likehoold function of model is defined by:

$$L(\mu, \sigma^2, Z_1, ...Z_n) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(Z_i - \mu)^2}{2\sigma^2}\right).$$

$$ln[L(\mu, \sigma^2, Z_1, ...Z_n)] = -\frac{n}{2}ln\sigma^2 - \frac{n}{2}ln(2\pi) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(Z_i - \mu)^2 \quad \text{with} \quad Z_i = \omega_l l_i + \omega_c c_i + b$$

$$ln[L(\mu, \sigma^2, b, Z_1, ...Z_n)] = -\frac{n}{2}ln\sigma^2 - \frac{n}{2}ln(2\pi) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(\omega_l l_i + \omega_c c_i + b - \mu)^2$$

$$\frac{\partial ln[L(\mu, \sigma^2, b, Z_1, ...Z_n)]}{\partial b} = 0 \Longleftrightarrow \sum_{i=1}^{n}(\omega_l l_i + \omega_c c_i + b - \mu) = 0$$

$$b = \mu - \omega_l \frac{\sum_{i=1}^{n} l_i}{n} - \omega_c \frac{\sum_{i=1}^{n} c_i}{n}$$

Using the weak law of large numbers,$E[\frac{\sum_{i=1}^{n} l_i}{n}] = \bar{l}$ and ,$E[\frac{\sum_{i=1}^{n} c_i}{n}] = \bar{c}$

$$b = \mu - \omega_l \bar{l} - \omega_c \bar{c}$$

,with $\mu = \bar{l}$, we obtain $b = (1 - \omega_l)\mu - \omega_c \sigma^2$

$$\text{In summarising}\textbf{(5.1.1.1)}, \begin{cases} \omega_l = \sqrt{ad} \\ \omega_c = \sqrt{be} \\ b = (1 - \omega_l)\mu - \omega_c \sigma^2 \end{cases}$$

5.1.2 Score of model:

To obtain the model score, our approach is the estimation of the proportion $\frac{\mu}{\sigma^2}$.Let's set the bias is equal to zero

$$b = 0 \Longleftrightarrow (1 - \omega_l)\mu - \omega_c \sigma^2 = 0$$

$$\frac{\mu}{\sigma^2} = \frac{\omega_c}{1 - \omega_l}$$

The score of the model is $\frac{\omega_c}{1-\omega_l}$

6 Results

To predict the evolution of dynamism over the years, we used the mathematical processing studied in [3].Knowing the populations at a given year t_i,our model was first used to predict the populations at a given year t_f.Then, taking into account the oscillatory behavior of the Volterra model and the periodicity T.We used the formulas $x(t) = \epsilon \cos(\omega t)$, $y(t) = -\frac{ae\epsilon}{b\omega} \sin(\omega t)$,to obtain the simulations (evolution of dynamism from t_i to t_f).

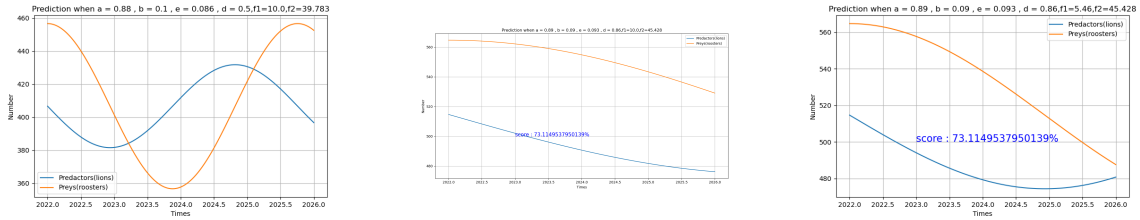


Figure3: Prediction based on an artificial neuron with valid scores

7 Discussions and perspectives

The values of w_1 and w_2 were determined through a mathematical treatment of the Volterra model. These values should maximize the likelihood of the artificial linear prediction model as much as possible. To verify this, one approach is to use the gradient descent method and then compare the results, including for the parameter b (the bias) in **(5.1.1.1)**.

The bias obtained through the maximum likelihood method must satisfy a positivity condition in machine learning,We find the condition as follows $\frac{\mu}{\sigma^2} \geq 1$. Regarding the model's score, it must be greater than 0.70 to be considered acceptable.Futhermore,the model'score is significantly if $\omega_l + \omega_c \leq 1$. The simulations confirm these conditions, although it is essential to properly calibrate the parameters to avoid chaotic behavior in the model.The neuron must be trained on large amounts of data. However, when the data size becomes too large, memory errors may occur. One approach to overcome this issue would be to use a supercomputer to perform large-scale predictions and enhance the performance of the artificial neuron.

An other approach to improving performance is to apply Hebb's theory as stated in **(5.1.1)**, to accomplish more complex tasks and refine predictions.

8 Conclusion

A first part of our work consisted in studying the Volterra model, which allowed us to design a Prey-Predator classification model using unsupervised learning and then predict the populations of prey and predators using an artificial neuron.

The model's score s depends on the parameters of the Lotka-Volterra model and is expressed as s when these parameters are properly calibrated. Our model serves as a foundation for understanding the dynamics of a Prey-Predator ecosystem, thus addressing a problem considered "classical" in artificial intelligence**[17]**.

By applying Hebb's theory **(5.1.1)**, we can construct multiple deep neural networks based on our neuron model to further analyze population dynamics beyond mere predation, exploring other ecological interactions such as symbiosis, commensalism, etc.

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